Supplementary Material: Residual TFT-EWC: Continual Learning for State-of-Health Forecasting and Reuse Screening of Heterogeneous Second-Life EV Batteries

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This supplement gives the preprocessing, the input correlations, the training curves and the full classification and regression metrics behind the paper. Every figure and table is generated by a script in the repository from committed results or from the public data; the script is named in each caption. SOH is relative to each cell's first valid capacity check.

S1. Preprocessing

Preprocessing is the same for every model and uses no label information beyond the SOH history at the origin.

1) *Parsing and verification.* Raw files are verified against their repository record or checksums. Our parsing reproduces the processed S1 RPT capacities and the authors' capacities of all 843 S6 checks exactly.
2) *Validity rules.* An S6 check is used only if its C/25 discharge starts from full charge and reaches the cut-off; 4 of 847 checks fail (Fig. S1c). In S1, RPT 2 (run at about 20 °C) is dropped.
3) *Noise reduction.* Incremental-capacity curves $-\mathrm{d}Q/\mathrm{d}V$ are computed on a 1 mV (S6) or 2 mV (S1) voltage grid after a Gaussian filter of the charge (5 mV in S6, 10 mV in S1); without it, differentiation amplifies the measurement noise (Fig. S1b). Peak positions and heights are taken from the smoothed curve; in S6 a peak at the edge of its window is set to missing. This is the only filtering; no other denoising or outlier removal is applied.
4) *Ageing cycles.* Apache Spark reduces 9.0×10^8 per-second rows of the S6 ageing cycles to block inputs (mean, last value and slope of seven responses per 100-cycle block).
5) *Inputs and scaling.* Features never observed are dropped (e.g. the second IC window in S6). Missing values are imputed with the training median and flagged; inputs are standardised. Imputation, scaling and target scaling are fitted on the first task and then frozen.

S2. Input correlations

Fig. S3 gives the Spearman correlation of the S6 history and diagnostic inputs (F1) with each other and with the target. Ten of the 17 diagnostics are near-monotone functions of the SOH at the origin ($|\rho| \geq 0.9$), so they add little beyond the SOH history; the fade rate is the input most correlated with the target ($\rho = 0.8$). This is consistent with the ablation in the paper, where removing the diagnostics improved the TFT.

S3. Training curves

Each task is trained with Adam (batches of 64) for at most 200 epochs, with early stopping on the task's validation RMSE (patience 20); the weights of the best epoch are kept. Fig. S4 shows one stream of the residual TFT + EWC (horizons 1–3, first evaluation order, seed 0, tuned learning rate $3 \cdot 10^{-3}$ and $\lambda = 1$). The re-run reproduces the committed final average error of that stream exactly (0.464 pp). Validation RMSE reaches its minimum within 2–30 epochs on five tasks and after 86 epochs on task 6; on task 5 no epoch improved on the weights carried over from task 4, which were therefore kept.

S4. Full metric sets (P2)

Tables SI and SII give every regression and classification metric of the P2 forecasts (decision D30); Fig. S5 gives the S6 confusion matrices and Fig. S6 the error per held-out group.

S5. Ageing-cycle processing and first-life pretraining

Spark job. Between checks, every S6 cell logged each ageing cycle at about one sample per second: 827 files, 132 GB raw and 9.0×10^8 rows. Apache Spark 4.2 (PySpark, local mode) reduces them to block-level inputs: one map task per 100-cycle block derives, for every cycle, a validity flag and seven measured responses (end-of-charge and end-of-discharge voltage, constant-voltage hold time and final current, relaxation after charge and after discharge, internal resistance); Spark SQL groups the cached per-cycle table (82,529 cycles) by cell and block into the mean, last value and slope of each response (21 inputs), and each block joins the origin at the check that ends it. Charge, energy and current are excluded because the protocol scales them with the previous capacity (within-cell $r = 0.97$–0.999 with the previous check's capacity). The run takes 20 min on six cores and matches an independent pandas implementation on all 827 blocks (largest difference 2×10^{-13}).

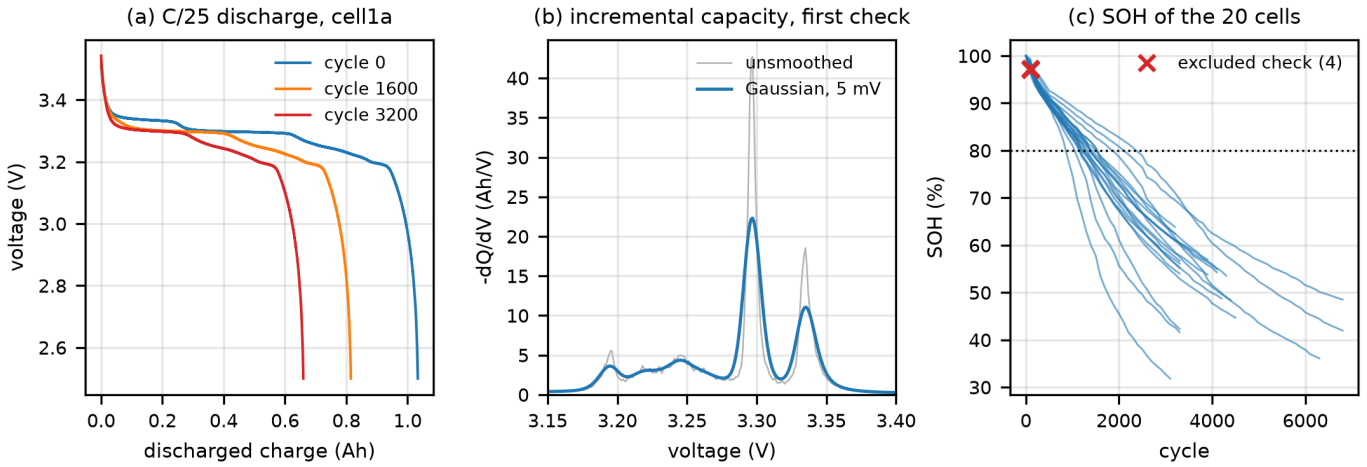


Fig. S1. S6 preprocessing (scripts/make_preprocessing_figures.py). (a) C/25 discharge of cell 1a at three ages. (b) Incremental capacity of its first check without and with the 5 mV Gaussian smoothing used for the IC features. (c) SOH of the 20 cells at every valid check; crosses: the four checks excluded by the validity rule, placed on the cell's curve at the cycle they were taken (they have no valid capacity). Dotted: 80% SOH.

TABLE SI

PROTOCOL P2, ALL REGRESSION METRICS OF THE FORECAST SOH (PP UNLESS STATED), MEAN OVER HELD-OUT GROUPS (THE RMSE COLUMN EQUALS TABLE P2 OF THE PAPER FOR THE BASELINES; MLP AND TFT ARE RE-RUNS ON ANOTHER PLATFORM, WITHIN 0.0052 PP OF IT). BIAS = MEAN(FORECAST − MEASURED): POSITIVE VALUES OVERESTIMATE SOH. NRMSE = RMSE / RANGE OF MEASURED SOH IN THE GROUP. R^2 IS ON THE SOH LEVEL, R^2_Δ ON THE SOH CHANGE SINCE THE ORIGIN (PERSISTENCE HAS $R^2_\Delta \leq 0$). SOH IS RELATIVE TO THE FIRST OBSERVED CHECK.

Data	Method	RMSE	MSE	MAE	MedAE	Bias	NRMSE	MAPE (%)	R^2	R^2_Δ
S1	Persistence	0.545	0.303	0.470	0.473	+0.430	0.161	0.481	0.672	-1.853
S1	Linear extrapolation	0.511	0.293	0.365	0.226	-0.014	0.148	0.374	0.690	-1.314
S1	Ridge, usage only	0.301	0.096	0.206	0.133	-0.038	0.088	0.212	0.897	0.181
S1	Ridge, F0	0.418	0.212	0.369	0.350	+0.205	0.118	0.379	0.798	-0.673
S1	Ridge, F1	0.320	0.115	0.266	0.241	+0.065	0.092	0.273	0.881	0.083
S1	Gradient boosting, F1	0.219	0.053	0.148	0.102	-0.021	0.064	0.152	0.944	0.543
S1	MLP	0.285	0.087	0.211	0.162	+0.009	0.083	0.216	0.907	0.260
S1	TFT	0.291	0.090	0.210	0.149	+0.027	0.085	0.215	0.904	0.216
S6	Persistence	2.817	8.389	2.451	2.225	+2.451	0.056	3.539	0.956	-3.160
S6	Linear extrapolation	1.372	1.946	0.738	0.333	+0.050	0.028	0.973	0.988	-0.050
S6	Ridge, usage only	1.177	1.607	0.864	0.671	-0.003	0.023	1.292	0.993	0.275
S6	Ridge, F0	0.859	0.905	0.613	0.439	+0.028	0.016	0.893	0.996	0.614
S6	Ridge, F1	1.137	1.767	0.749	0.550	+0.013	0.022	1.130	0.991	0.221
S6	Gradient boosting, F1	0.710	0.725	0.488	0.321	+0.097	0.013	0.710	0.997	0.736
S6	MLP	0.826	1.018	0.608	0.444	+0.220	0.015	0.914	0.996	0.656
S6	TFT	0.868	1.151	0.621	0.424	+0.235	0.016	0.928	0.996	0.622

First-life pretraining (P3). The pool provides 36,711 samples, against 2,409 in S6, and validation selected EWC $\lambda = 0.01$ for the pretrained TFT. Under the replication rule, EWC and fine-tuning are practically equivalent for the pretrained TFT (differences of 0.015–0.017 pp). The S6 stream does not degrade the pool test error (TFT + EWC: 3.280 to 3.148 pp), but on that split the pretrained models are also less accurate than extrapolation (3.001 pp): pretraining does not learn a first-life model worth transferring.

Attention. The TFT's attention over the four previous origins is nearly uniform (0.24–0.26 per lag): it makes little use of its history.

S6. ADDITIONAL RESULTS

Tables SIII and SVI and Figs. S7 and S8 belong to Sections IV-D to IV-G of the paper and were moved here for space.

Reproducibility of the comparisons (Section IV-B). P1 and P2 were re-run in a pinned environment on data rebuilt from the raw archives. Every baseline, every P2 result and the ranking of methods reproduced; the neural models moved by at most 0.059 pp. On S1, five Holm-corrected comparisons with TFT + EWC crossed $p = 0.05$. On S6, validation selected EWC $\lambda = 1000$ instead of 100 (a 0.008 pp tie), and the error advantage of TFT + EWC over fine-tuning moved from $p_{\text{Holm}} = 0.005$ to 0.060, while its reduction of forgetting moved from 0.050 to 0.008. These findings motivated the ten tuning orders, the one-standard-error rule and the replication rule used from P3 on. A D30 re-run of P2 on another platform reproduced every baseline exactly and moved the MLP and TFT by at most 0.027 pp per held-out group: neural training is deterministic per platform, not across platforms.

Reuse classes in the D30 re-run of P2. On S6 the learned models assign the measured reuse class to 95–97% of samples

TABLE SII

PROTOCOL P2, REUSE CLASS (C0–C3, D10 CUT-POINTS) AND END-OF-FIRST-LIFE DECISION (SOH $\leq$ 80, POSITIVE) OF THE FORECAST SOH AGAINST THE MEASURED SOH, ALL HELD-OUT SAMPLES POOLED (MEAN OVER SEEDS). MACRO AVERAGES RUN OVER THE MEASURED CLASSES. AUC USES THE FORECAST SOH AS THE SCORE (POINT FORECASTS; NO CLASS PROBABILITIES, SO NO LOG LOSS). MEASURED CLASS COUNTS OF ALL SAMPLES: S1: C0 279, C1 0, C2 0, C3 0; S6: C0 52, C1 182, C2 516, C3 1659. – : UNDEFINED (ONE CLASS ONLY).

Data	Method	Acc.	Bal. acc.	Prec.	Rec.	Spec.	F1	MCC	κ	κ_w	EoL Prec.	EoL Rec.	EoL Spec.	EoL F1	EoL MCC	EoL AUC
S1	Persistence	1.00	1.00	1.00	1.00	–	1.00	–	–	–	–	–	1.00	–	–	–
S1	Linear extrapolation	0.98	0.98	1.00	0.98	–	0.99	–	0.00	0.00	–	–	1.00	–	–	–
S1	Ridge, usage only	1.00	1.00	1.00	1.00	–	1.00	–	0.00	0.00	–	–	1.00	–	–	–
S1	Ridge, F0	1.00	1.00	1.00	1.00	–	1.00	–	–	–	–	–	1.00	–	–	–
S1	Ridge, F1	1.00	1.00	1.00	1.00	–	1.00	–	–	–	–	–	1.00	–	–	–
S1	Gradient boosting, F1	0.99	0.99	1.00	0.99	–	0.99	–	0.00	0.00	–	–	1.00	–	–	–
S1	MLP	1.00	1.00	1.00	1.00	–	1.00	–	0.00	0.00	–	–	1.00	–	–	–
S1	TFT	1.00	1.00	1.00	1.00	–	1.00	–	0.00	0.00	–	–	1.00	–	–	–
S6	Persistence	0.86	0.78	0.62	0.78	0.96	0.66	0.72	0.72	0.81	1.00	0.93	1.00	0.96	0.89	1.00
S6	Linear extrapolation	0.94	0.84	0.81	0.84	0.98	0.78	0.88	0.87	0.91	1.00	0.98	0.99	0.99	0.96	1.00
S6	Ridge, usage only	0.95	0.91	0.83	0.91	0.98	0.85	0.90	0.90	0.93	0.99	0.98	0.98	0.99	0.96	1.00
S6	Ridge, F0	0.96	0.88	0.89	0.88	0.99	0.88	0.92	0.92	0.94	1.00	0.98	0.99	0.99	0.97	1.00
S6	Ridge, F1	0.96	0.84	0.88	0.84	0.98	0.85	0.91	0.91	0.93	0.99	0.99	0.97	0.99	0.96	1.00
S6	Gradient boosting, F1	0.97	0.89	0.89	0.89	0.99	0.89	0.93	0.93	0.95	0.99	0.99	0.98	0.99	0.97	1.00
S6	MLP	0.96	0.86	0.88	0.86	0.99	0.87	0.92	0.92	0.94	0.99	0.99	0.98	0.99	0.97	1.00
S6	TFT	0.96	0.88	0.89	0.88	0.99	0.88	0.92	0.92	0.94	0.99	0.98	0.99	0.99	0.96	1.00

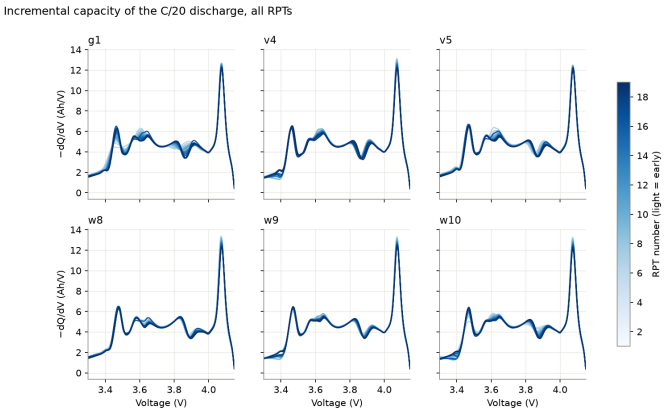


Fig. S2. S1: incremental-capacity curves of the C/20 RPTs (10 mV smoothing), notebook 11.

TABLE SIII

LEARNED RESIDUAL, SHIFT-ADAPTIVE EWC AND LITERATURE BASELINES ON S6 (STREAM FROM SCRATCH): FINAL AVERAGE TEST RMSE (PP), MEAN OVER 50 RUNS, MAIN / REPLICATE SEED SETS, AND FORGETTING (PP) FOR HORIZONS 1–6. BELOW THE LINE: SINGLE FITS ON ALL TASKS (NOT CONTINUAL) AND EXTRAPOLATION.

Method	E1: h=1–3	E2: h=1–6	Forgetting, E2
TFT + EWC, residual	**0.339 / 0.314**	**0.561 / 0.534**	**0.093 / 0.086**
TFT + SA-EWC, residual	0.343 / 0.321	0.548 / 0.564	0.066 / 0.084
TFT fine-tuning, residual	0.347 / 0.337	0.673 / 0.612	0.282 / 0.220
TFT + EWC, direct	0.504 / 0.487	0.802 / 0.835	0.135 / 0.166
MLP + EWC, residual	0.374 / 0.366	0.659 / 0.694	0.205 / 0.225
BiLSTM + EWC, residual	0.328 / 0.370	0.669 / 0.765	0.229 / 0.318
Transformer + EWC, residual	0.360 / 0.366	0.609 / 0.705	0.129 / 0.198
TCN + EWC, residual	0.466 / 0.408	0.812 / 0.756	0.345 / 0.268
Linear extrapolation	0.343	0.485	–
Gradient boosting	0.451	0.898	–
SVR, residual	0.429	0.616	–
GPR, direct	1.374	2.073	–

(MCC 0.90–0.93), against 94% (0.88) for linear extrapolation and 86% (0.72) for persistence, and all errors but one (persistence) are between neighbouring classes; the end-of-first-life decision does not separate the methods (AUC $\geq$0.998), as 69% of S6 samples are below 80%.

Intervals on unseen conditions. Intervals (Fig. S8) calibrated on the pooled validation rows of the training conditions undercover on the held-out one, for the TFT (pooled 0.66–0.68) and for extrapolation (0.74–0.76) alike. Calibrating extrapolation, which has no fitted parameters, on all origins of those conditions instead of their mid-life validation rows restores 0.89–0.90: the calibration rows caused the shortfall, not only the model. With leave-one-condition-out calibration, gradient-boosting intervals reach the registered threshold of 0.85 (0.90 in both horizon sets) with the lowest mean interval scores (3.2 and 6.7 pp, against 7.1 and 11.8 pp for calibrated extrapolation and 6.1–6.3 and 10.0–10.3 pp for the TFT), lower than calibrated extrapolation's on six of seven conditions (significant only for horizons 1–3, $p = 0.047$, against $p = 0.30$, so not established). Coverage holds over conditions, not for each: on the 100% depth-of-discharge condition, outside the range of the others, it is 0.48 and 0.38.

Table SV lists the shrunk residual TFT + EWC; the raw correction gives 0.88/0.90, 0.79/0.80, 1.59/1.62 and 1.46/1.49 pp in the same columns.

Gradient boosting under the continual constraint (D32). Table SIV is the first P1 benchmark (Section IV-A of the paper). The headline comparison of the paper (its Table III) adds gradient boosting trained like the proposed model on unseen conditions: the six training conditions arrive one at a time and earlier conditions' data are not kept. Refitted on the latest condition only, it reaches 0.86 and 1.59 pp (horizons 1–3 and 1–6), against 0.92/0.94 and 1.62/1.65 pp for the proposed model (slightly better for 1–3, $+0.06/+0.08$ pp, replicated; not replicated for 1–6) and 0.54 and 1.05 pp with all six conditions stored. Warm-started condition by condition (200 more trees per condition), it reaches 2.65 and 3.49 pp, and the proposed model is better by 1.7–1.9 pp (replicated). This gap

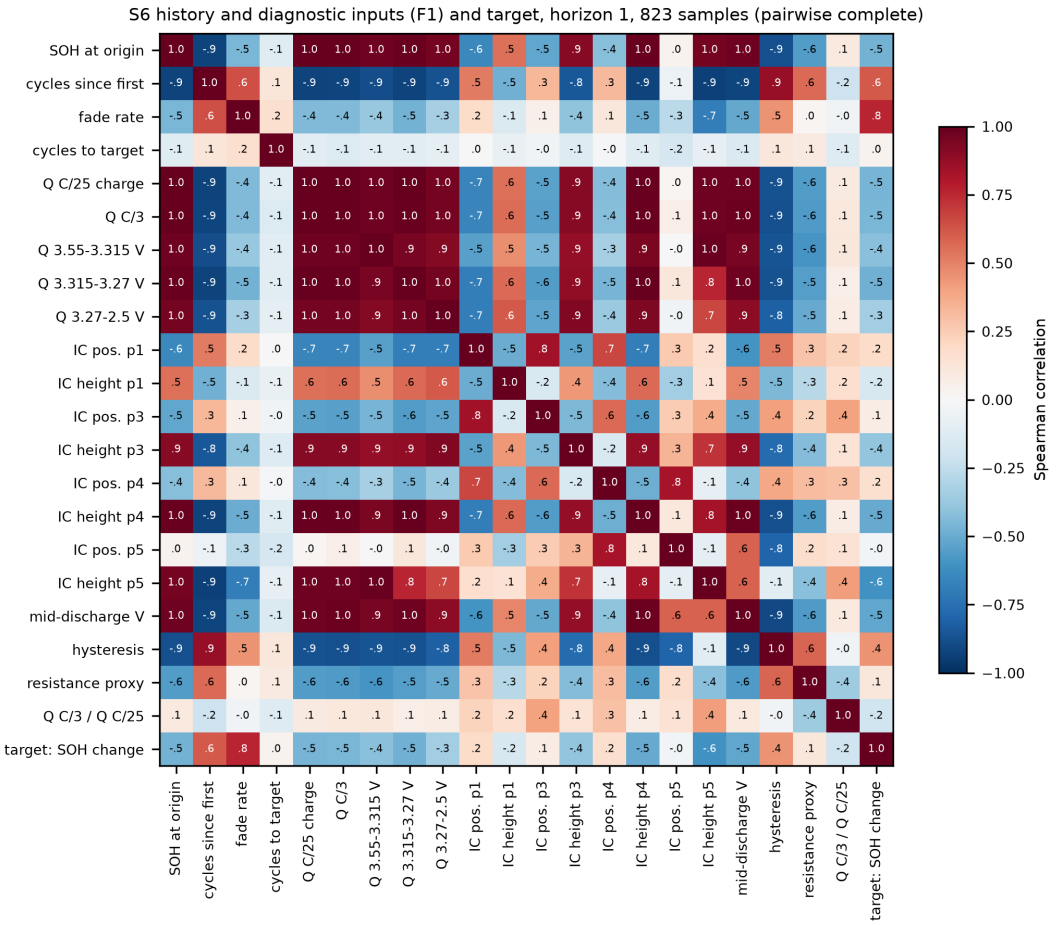


Fig. S3. Spearman correlation of the S6 F1 inputs and the target Δ (horizon 1, pairwise complete samples; the static test conditions are omitted). scripts/make_preprocessing_figures.py.

is mostly an artefact: scikit-learn keeps the bin edges of the first condition when warm-starting, and every usage input is constant within a condition, so the later trees cannot use them. We therefore do not count it as evidence against continual gradient boosting in general.

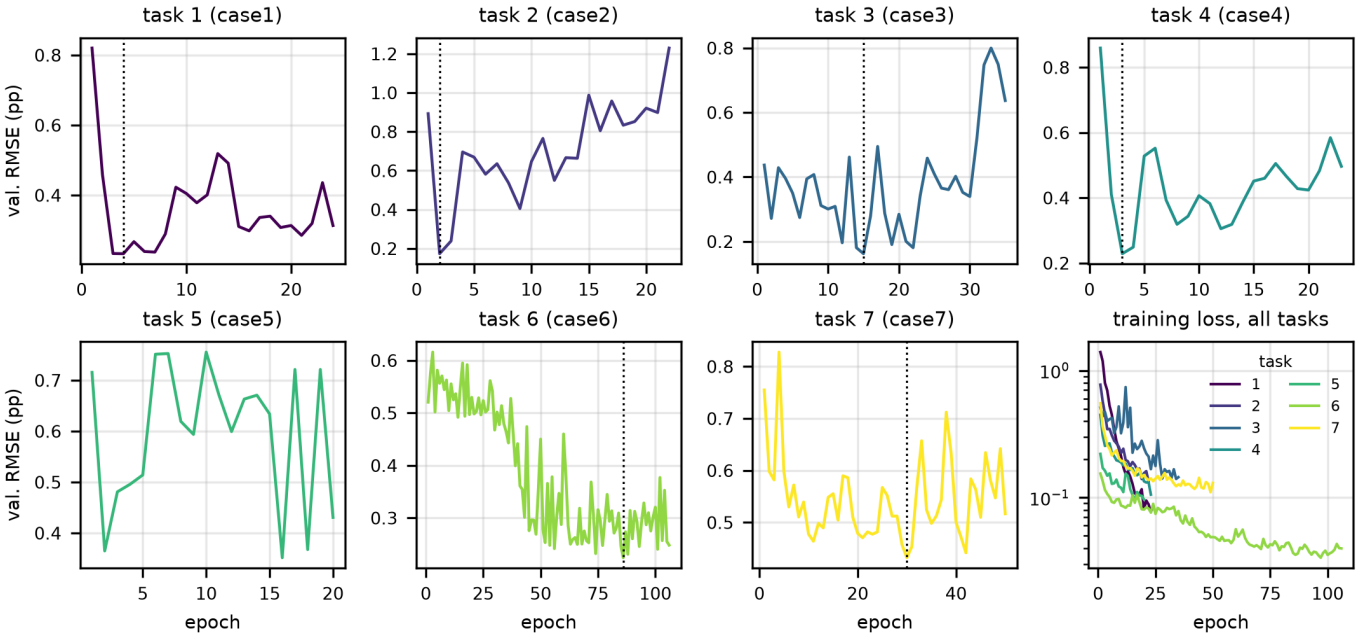


Fig. S4. Residual TFT + EWC on the S6 stream: validation RMSE per epoch for each task (dotted: early-stopping epoch) and training loss (scaled MSE plus EWC penalty) of all tasks. scripts/run_epoch_curves.py, results/epoch_curves_d23_e1.json.

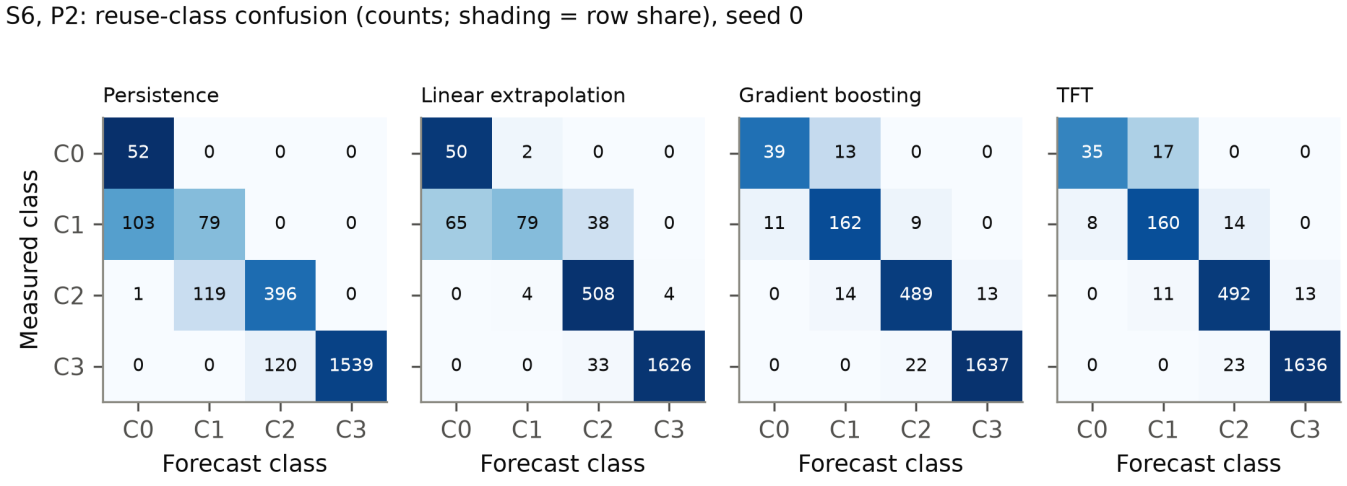


Fig. S5. S6, P2: reuse-class confusion matrices (counts; shading: row share), seed 0. scripts/make_d30_outputs.py.

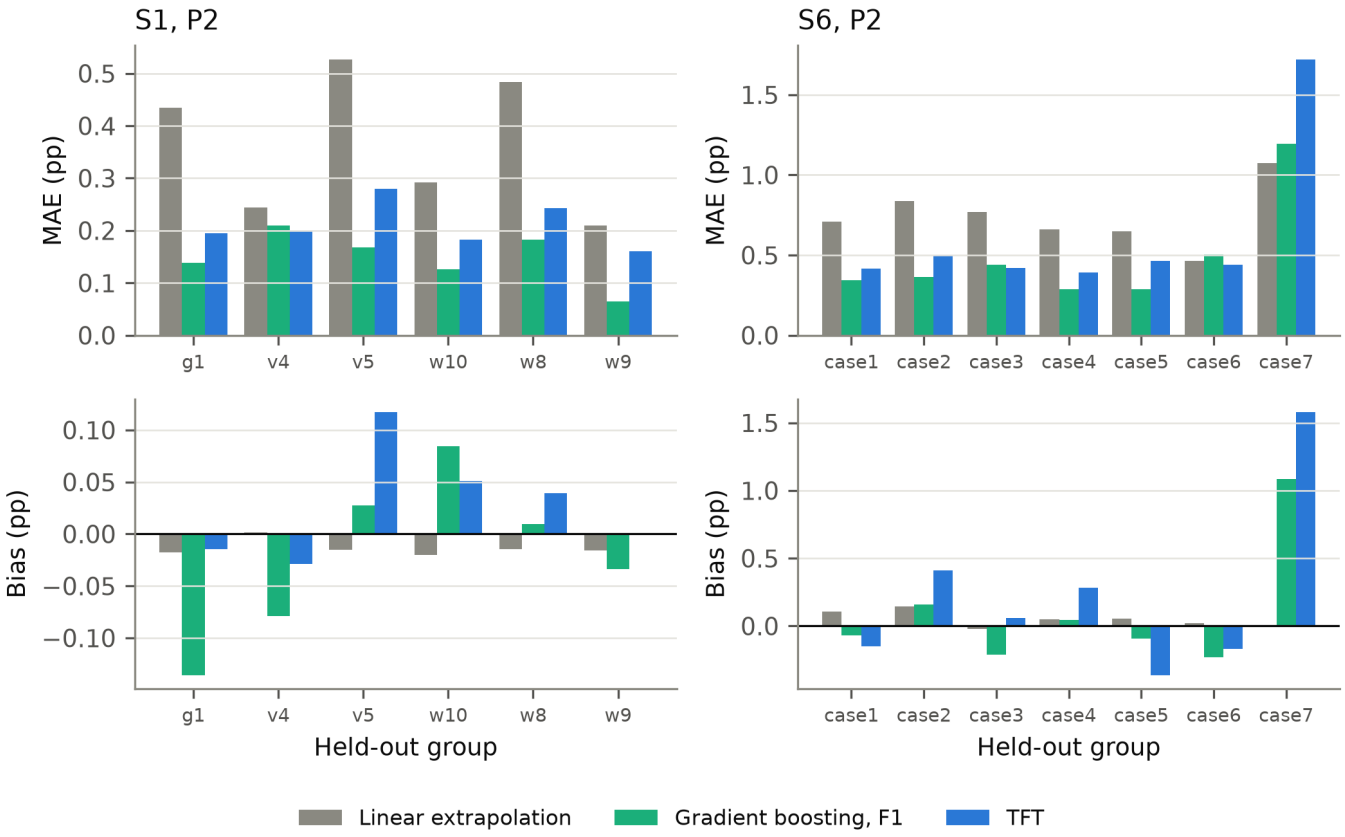


Fig. S6. P2: error per held-out group. scripts/make_d30_outputs.py.

TABLE SIV

PROTOCOL P1 (CONTINUAL STREAM). FINAL AVERAGE TEST RMSE OF SOH($k+h$) IN PP (MEAN ± S.D. OVER 10 ORDERS × 5 SEEDS FOR CONTINUAL METHODS, OVER 5 SEEDS FOR JOINT TRAINING), SKILL RELATIVE TO PERSISTENCE ($1 - \mathrm{RMSE}/\mathrm{RMSE}_{\mathrm{persistence}}$) AND FORGETTING (PP). †UPPER BOUND, TRAINED ON ALL TASKS AT ONCE.

Method	S1: six NMC second-life cells RMSE	Skill	Forgetting	S6: 20 LFP cells, seven usage conditions RMSE	Skill	Forgetting
Persistence	0.632	0.00	–	1.644	0.00	–
Linear extrapolation	0.726	-0.15	–	0.343	0.79	–
Ridge, usage only	0.450	0.29	–	1.274	0.22	–
Ridge, F0	0.678	-0.07	–	0.801	0.51	–
Ridge, F1	0.491	0.22	–	0.605	0.63	–
Gradient boosting, F1	0.466	0.26	–	0.372	0.77	–
MLP fine-tuning	0.492 ± 0.029	0.22	0.030	0.531 ± 0.151	0.68	0.207
MLP replay	0.475 ± 0.021	0.25	0.025	0.597 ± 0.178	0.64	0.212
MLP + EWC	0.497 ± 0.033	0.21	0.040	0.540 ± 0.176	0.67	0.212
MLP joint†	0.500 ± 0.016	0.21	–	0.414 ± 0.037	0.75	–
TFT fine-tuning	0.467 ± 0.014	0.26	0.026	0.883 ± 0.538	0.46	0.461
TFT + EWC	0.462 ± 0.019	0.27	0.016	0.677 ± 0.261	0.59	0.274
TFT + EWC, F0	0.472 ± 0.020	0.25	0.019	0.621 ± 0.257	0.62	0.176
TFT joint†	0.455 ± 0.017	0.28	–	0.427 ± 0.049	0.74	–

TABLE SV

UNSEEN S6 USAGE CONDITION (P2, SEVEN HELD-OUT CONDITIONS): TEST RMSE (PP), MEAN OVER FOLDS, ON ALL ORIGINS AND ON ORIGINS WITH A FADE RATE (H^+). TFT ROWS: MEAN OVER 50 RUNS, MAIN / REPLICATE SEED SETS; OTHER ROWS: SINGLE DETERMINISTIC FITS.

Method	E1: h=1–3 all	E1: h=1–3 H^+	E2: h=1–6 all	E2: h=1–6 H^+
TFT + EWC, residual, shrunk	0.92 / 0.94	0.74 / 0.74	1.62 / 1.65	1.42 / 1.44
TFT, residual, joint	0.57 / 0.58	0.54 / 0.55	1.21 / 1.18	1.22 / 1.19
Gradient boosting	**0.54**	**0.51**	**1.05**	**1.05**
Ridge	0.77	0.69	1.30	1.26
Linear extrapolation	1.37	0.86	2.30	1.62
Persistence	2.82	2.66	4.90	4.73

TABLE SVI

REUSE CLASS OF THE FORECAST SOH ON UNSEEN S6 CONDITIONS (P2, HORIZONS 1–3, SEVEN HELD-OUT CONDITIONS POOLED; MEASURED C0 52, C1 182, C2 516, C3 1659). SENS., PREC.: MACRO RECALL AND PRECISION. PROPOSED MODEL: MEAN AND S.D. OVER 50 RUNS; OTHERS: SINGLE FITS. BOLD: BEST.

Approach	Acc.	Sens.	Prec.	F1	MCC
Persistence	0.858	0.782	0.625	0.661	0.723
Linear extrapolation	0.939	0.840	0.810	0.777	0.876
Ridge	0.965	0.857	0.861	0.858	0.927
Gradient boosting	**0.977**	**0.924**	**0.926**	**0.925**	**0.952**
Residual TFT-EWC (proposed)	0.955	0.880	0.840	0.840	0.906
s.d. over 50 runs	±0.005	±0.017	±0.016	±0.021	±0.011

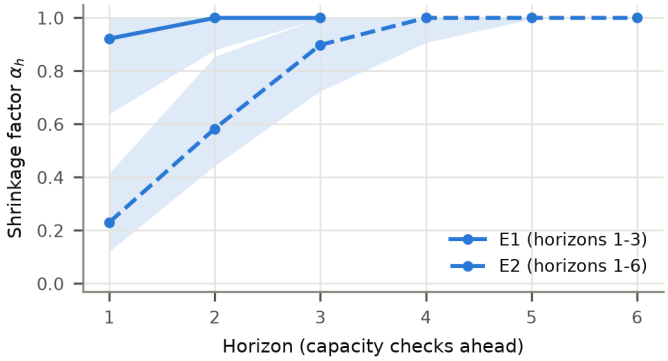


Fig. S7. S6 stream: shrinkage factor α_h of the TFT correction per horizon after the last task (median and interquartile range over 50 runs, seeds 0–4); $\alpha_h = 0$ is linear extrapolation.

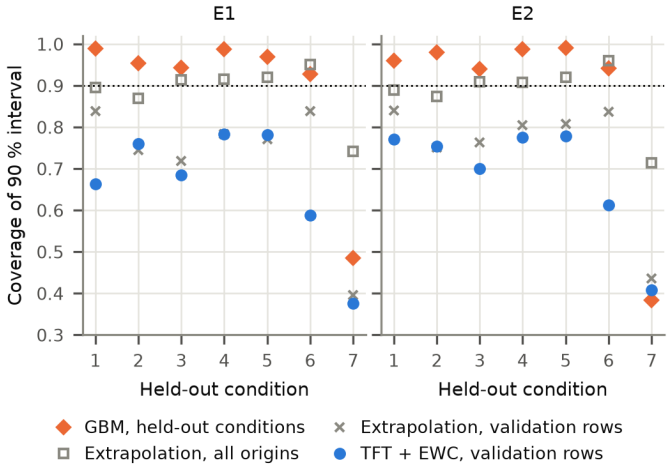


Fig. S8. Unseen S6 conditions: coverage of 90% conformal intervals per held-out condition, by forecaster and calibration rows. TFT: mean over 50 runs (seeds 0–4). Dotted: nominal level.